

Name	Subject Class	Class	Candidate Number
	2BI		



ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2017

BIOLOGY

HIGHER 2

9744/02
17 AUGUST 2017
2 hours

Paper 2

READ THESE INSTRUCTIONS FIRST

Write your name, index number and class on this answer booklet.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
5	
6	
7	
8	
9	
Total	100

This question paper consists of **22** printed pages.

1 Fig. 1.1 shows part of a cell.

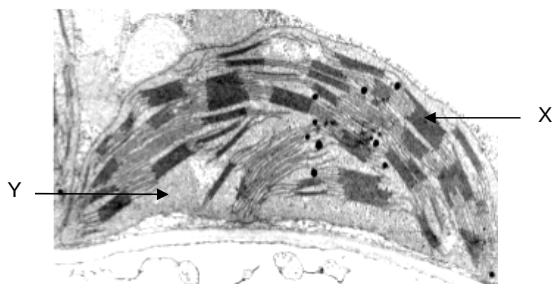


Fig. 1.1

(a) (i) Outline the role of the organelle labelled X.

[2]

(ii) Explain the importance of the double membrane enclosing organelle X for the reactions that occur in the region labelled Y.

[2]

(b) (i) Plant cells are unable to carry out endocytosis due to the energy needed to overcome the turgor pressure in the plant cell. However, lysosome-like organelles can be found in the plant cells. Suggest reasons for their presence in plant cells.

[2]

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(ii) Outline how the lysosome is formed by the endomembrane system in a eukaryotic cell.

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[3]

The chloroplast is a member of a larger family of plant organelles called plastids. All plastids, including chloroplasts, develop from proplastids. Fig. 1.2 shows the process of chloroplast development from a proplastid.

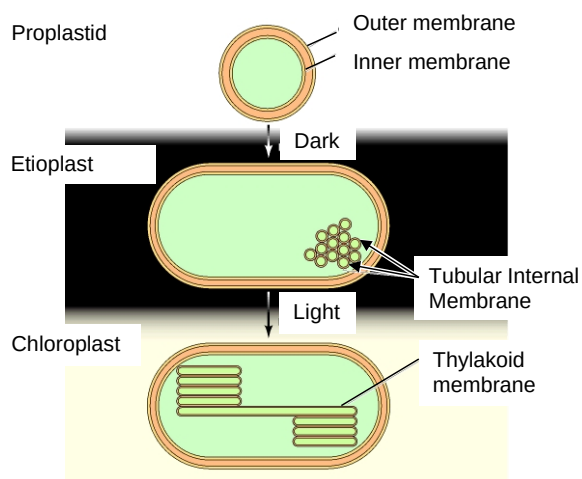


Fig. 1.2

(c) With reference to Fig. 1.2, suggest how thylakoid membranes are formed.

[2]

[Total: 11 m]

2 Fig. 2.1 shows the formation of a bond in the synthesis of starch.

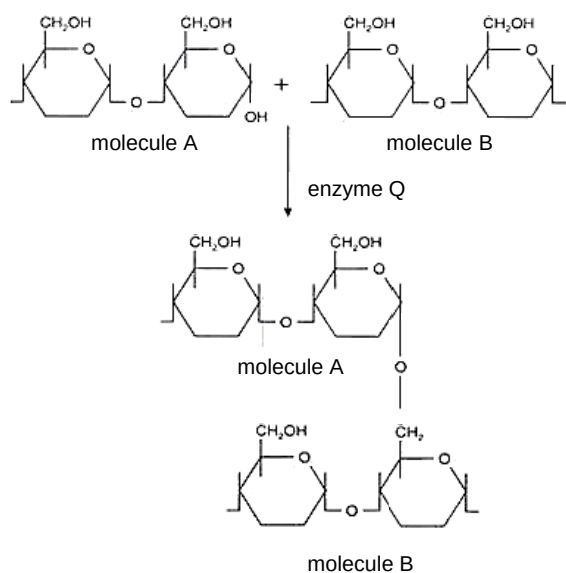


Fig. 2.1

(a) (i) Describe the formation of the bond in Fig. 2.1.

[2]

(ii) Explain how enzyme Q could lower the activation energy of the bond formation.

[3]

Graph X shows the amount of product formed over time for an enzyme-catalysed reaction (Fig. 2.2).

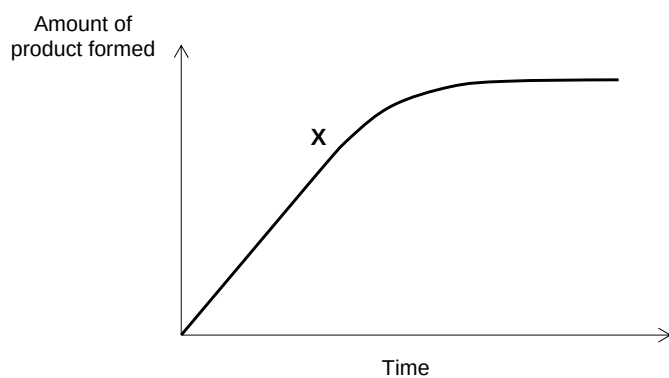


Fig. 2.2

(b) (i) Draw a graph, labelled Y, on Fig. 2.2 to show how a small amount of non-competitive inhibitor affects the amount of product formed over time. [1]

(ii) Explain the difference between graphs X and Y.

[2]

[Total: 8 m]

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- 3 The diagram below shows the structure of a mature tRNA for the amino acid alanine.

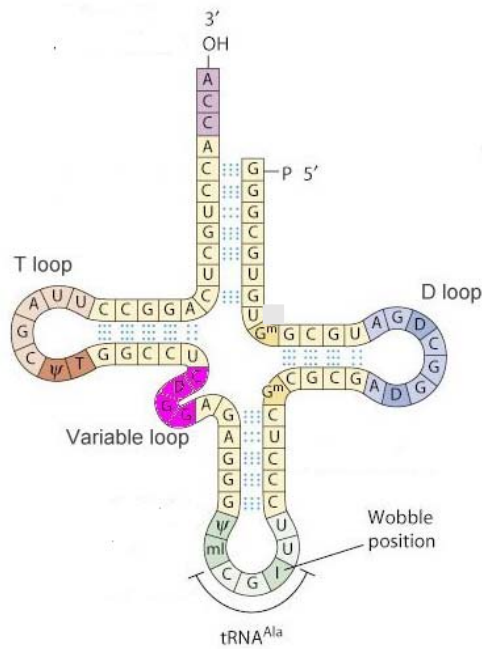


Fig. 3.1

- (a) (i) Explain the roles of hydrogen bonds in the proper functioning of tRNA.

[4]

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- (ii) The length of the tRNA gene is longer than that of the mature tRNA. Outline how a tRNA molecule is synthesised in eukaryotes.

[3]

An experiment was carried out to investigate the effects of various cytokines on a culture of CD34 cells. CD34 cells are hematopoietic progenitor cells which are produced in the early stage of differentiation to form mature immune cells from stem cells. Three types of cytokines, KF36EG, K36EG and F36EG, were used in the experiment as shown in Fig. 3.2.

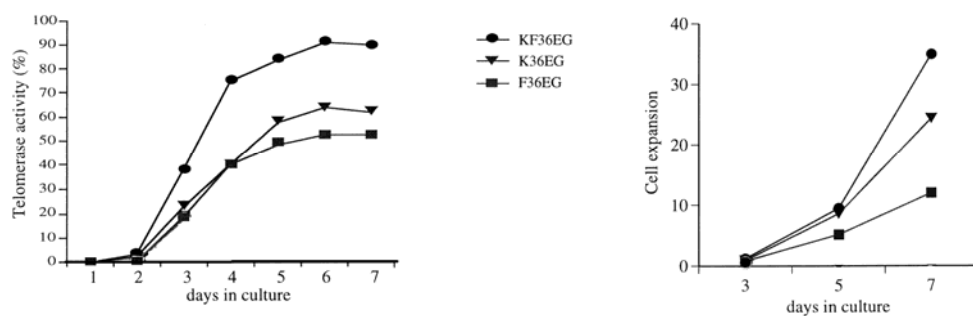


Fig. 3.2

- (b) (i) With reference to Fig. 3.2, describe the effects of the types of cytokines on CD34 cells.

[3]

- (ii) With reference to Fig. 3.2, comment on the role of cytokines on the CD34 cells.

[2]

- (c) Discuss the accuracy of the following statement:

“Stem cells and cancer cells are able to divide indefinitely as there is no end replication problem due to the presence of active telomerase.”

[4]

[Total: 16 m]

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- 4 The coat colour of Norwegian cattle is mainly determined by the distribution of two pigments: red and black. Both pigments are produced by the action of the enzyme tyrosinase in cells called melanocytes. Low enzyme activity leads to the production of red pigment, while high enzyme activity brings about black pigment production.

The activity of the enzyme is increased when melanocyte stimulating hormone (MSH) combines with a MSH receptor. The receptor is coded for by the gene, **R**, which has three alleles, **R^D**, **R^A** and **r**. **R^D** and **R^A** each codes for a receptor with a different activity. No receptor is produced by the recessive allele, **r**.

The dominant allele of a second gene, **B**, codes for a protein which binds to and blocks the MSH receptors coded for by **R^A**, thus preventing stimulation of tyrosinase activity in a melanocyte. The receptors coded for by **R^D** is insensitive to the protein coded by **B**. The recessive allele, **b**, does not produce a functional protein.

- (a) (i) State the name given to the interaction between the **R** and **B** gene loci.

_____. [1]

- (ii) Explain why animals with the genotype **R^AR^ABB** have red coats.

_____. [1]

- (iii) A red cow, with genotype **R^AR^ABB** is mated with a bull which is homozygous recessive at both gene loci. Draw a genetic diagram in the space below to show the expected genotypes and phenotypes and their ratios in the **F₁** and **F₂** generations.

[5]

During a health screening exercise of cattle in a farm, the height of the bulls was measured and the data collected is shown in Table 4.1.

Table 4.1

Height/cm	Number of bulls
131—135	3
136—140	9
141—145	21
146—150	12
151—155	2

- (b) Distinguish between the two types of variation as seen in coat colour and height in the Norwegian cattle.

[3]

In the honey bee colony, the queen bee is solely responsible for laying eggs and the drones for fertilizing her. The worker bees have well-developed mouthparts and structural adaptations for collecting nectar and pollen to gather food and to perform other duties in the hive. Male bees are developed from haploid eggs while both queen and worker bees develop from fertilized eggs.

- (c) Explain how the phenotypic differences between the queen and the worker bees come about despite both being developed from fertilized eggs.

[2]

[Total: 12 m]

- 5 Normal cells rely on oxidative phosphorylation in the mitochondria to generate the energy needed for cellular processes. In contrast, cancer cells undergo a phenomenon termed the "Warburg effect". This is characterised by an increased glucose uptake and reliance on glycolysis for ATP production despite the availability of oxygen for oxidative phosphorylation. Most of the pyruvate is converted to lactate instead of being broken down in the mitochondria (Fig. 5.1).

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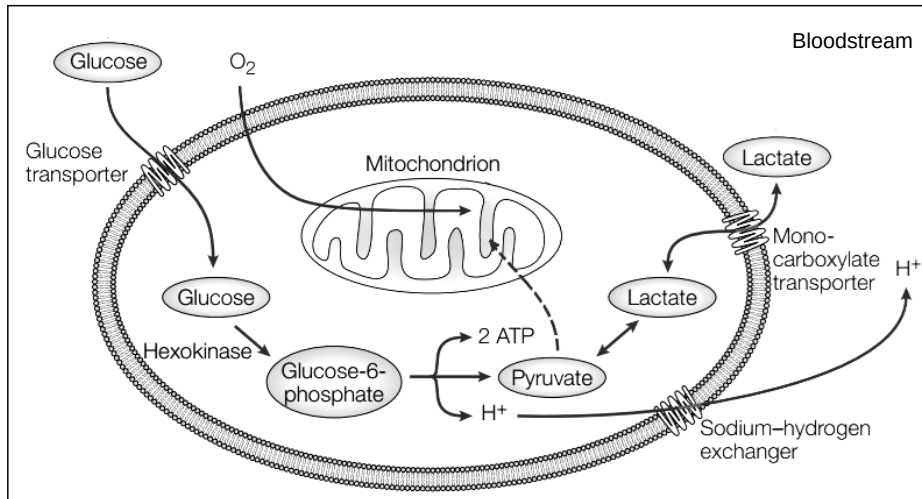


Fig. 5.1

- (a) (i) Outline the role of glycolysis in normal cells.

[3]

- (ii) With reference to Fig 5.1 and your own knowledge, suggest why the "Warburg effect" may seem disadvantageous to the survival of the cancer cell.

[1]

In cancer cells, the “Warburg effect” is constitutively upregulated even under normal levels of oxygen. It is thought to be due to the reprogramming of metabolic genes to increase glucose consumption. In a research experiment, the mean glucose consumption rate of MCF-7 breast cancer cells is compared with that of the relatively more aggressive MDA-MB-231 breast cancer cells under normal and low oxygen levels (Fig. 5.2).

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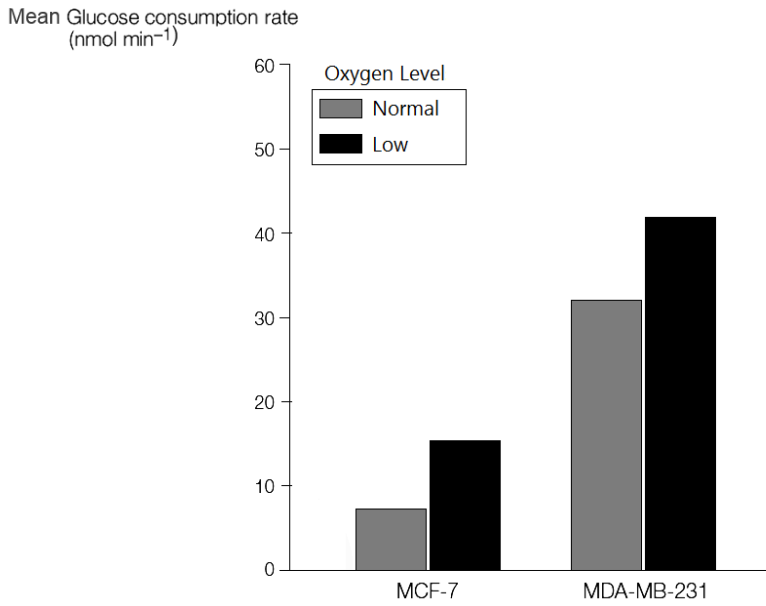


Fig. 5.2

- (b) Describe the differences in the mean glucose consumption rate between MCF-7 and MDA-MB-231 cancer cells.

[3]

The mTOR intracellular signalling pathway is critical for control of cell growth. Fig. 5.3 shows the signalling system that drives cell growth through greatly stimulating glucose uptake and utilisation in a normal cell.

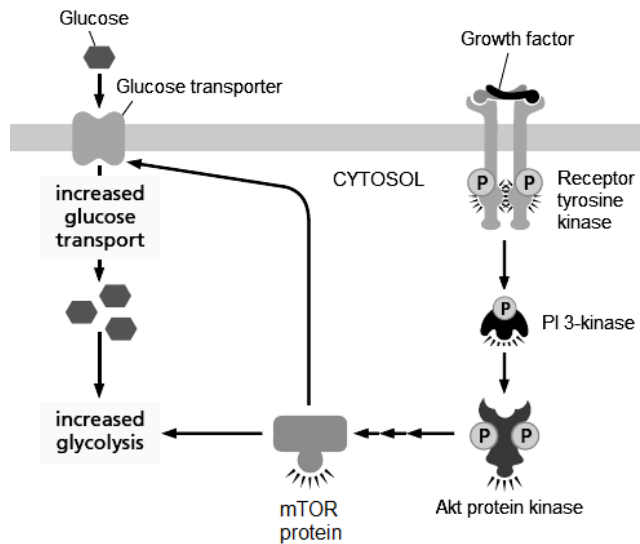


Fig. 5.3

- (c) (i) Describe how the binding of the growth factor to the receptor tyrosine kinase leads to the cellular responses in Fig. 5.3.

[3]

- (ii) Cancer cells are able to proliferate in the absence of growth factors. Suggest how the ability to do so can lead to the "Warburg effect".

[1]

[Total: 11 m]

6 Fig. 6.1 shows the different stages of meiosis in a cell.

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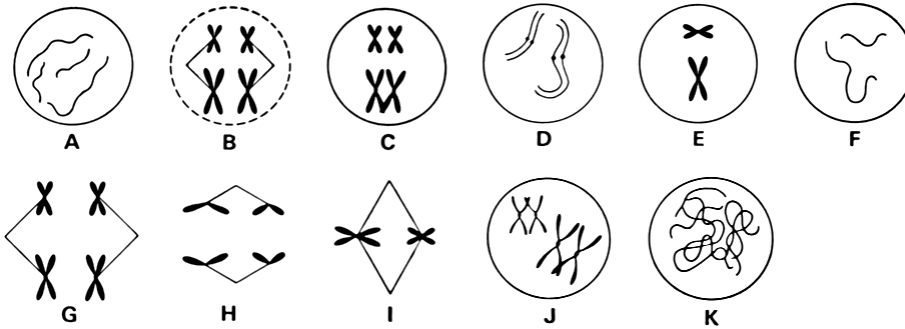


Fig. 6.1

(a) (i) Arrange the stages in chronological order.

[1]

		D			B					
--	--	---	--	--	---	--	--	--	--	--

(ii) With reference to Fig. 6.1, identify a stage which is critical for a reduction division and explain why.

[2]

(iii) Explain how reduction division contributes to variation in diploid organisms.

[2]

15

Some patients with the Prader-Willi syndrome have both chromosomes 15 from their mother due to an error in meiosis. Fig. 6.2 shows the possible combinations of chromosomes 15 that could be present in the gametes of the mother.

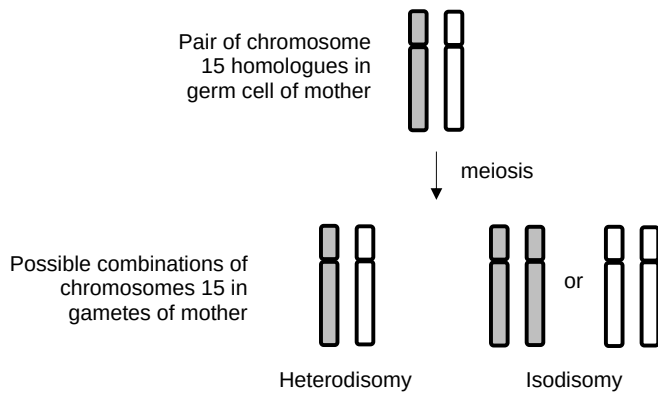


Fig. 6.2

(b) Explain how a gamete with heterodisomy of chromosome 15 could have been formed.

[3]

[Total: 8 m]

- 7 The structure of a G-protein-linked receptor (GPLR) is shown from two different views in Fig. 7.1 below. Fig. 7.1a shows a GPLR from a cross section of the plasma membrane while Fig. 7.1b shows the top view of a GPLR.

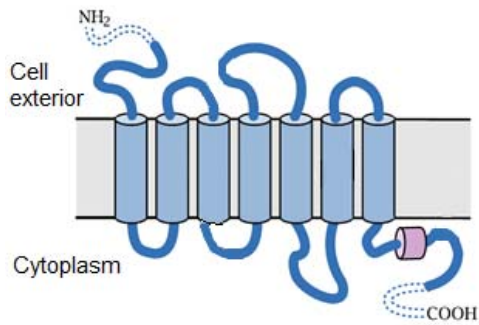


Fig 7.1a

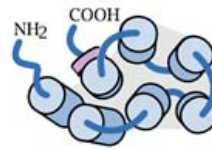


Fig 7.1b

- (a) (i) State and explain the highest level of organization of the GPLR.

[2]

- (ii) Identify a hydrophilic domain of the GPLR and describe its corresponding function.

[2]

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Recent studies have found that GPLRs have alternative binding sites other than the ligand-binding sites. Allosteric molecules bind to these alternative binding sites to change cellular response levels. One such molecule has been found to bind to the GPLR specific to glucagon.

One of the normal cellular responses of glucagon binding is the inhibition of glycogenesis. The rate of glycogenesis with and without the allosteric molecule is shown in Fig. 7.2.

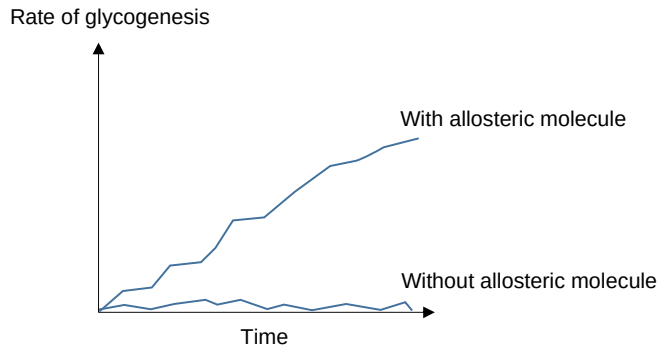


Fig. 7.2

- (b) (i) Based on the information above, state if the allosteric molecule is an activator or an inhibitor of GPLR.

[1]

- (ii) Explain the effect of the allosteric molecule on the rate of glycogenesis.

[5]

[Total: 10 m]

- 8 Speciation events have been observed to occur very frequently in bacteria. It was suggested that the high rate of speciation is due to the high level of variation in bacteria.

- (a) Transformation and conjugation are two processes which increase the level of variation in bacteria.

Distinguish these two processes.

[3]

Bacterial evolution is one of the most dynamic and exciting areas in current biological research.

Over the years, a barrier in this field of research is the difficulty in classifying bacterial species. However, in recent times, new analytical tools in molecular biology have offered new insights into the classification of bacterial species.

- (b) (i) Suggest why scientists had difficulties in the classification of bacterial species.

[2]

- (ii) Explain how analytical molecular tools have helped overcome this barrier in research.

[2]

(c) The outer layers of the two types of bacteria with peptidoglycan cell walls known as Gram-positive and Gram-negative bacteria are shown in Fig. 8.1 below.

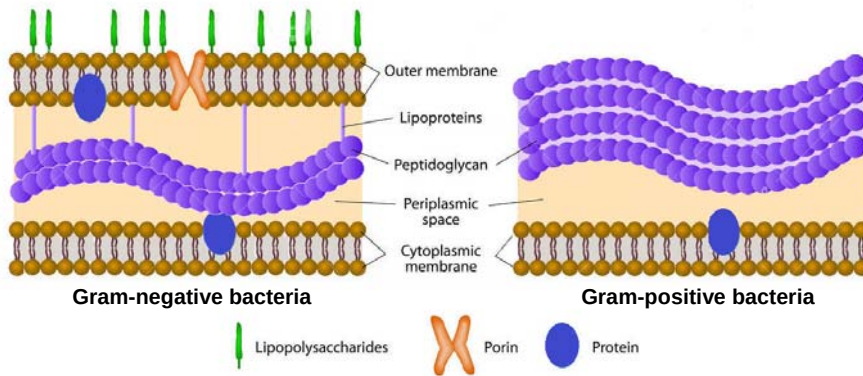


Fig. 8.1

Penicillin is an antibiotic that is known to be effective against only one of the two types of bacteria above.

With reference to the information given and your own knowledge, deduce which type of bacteria is susceptible to the action of penicillin and explain why.

[3]

- (d) Explain how antibiotic-resistant bacteria can become increasingly common in a population of bacteria.

[4]

[Total: 14 m]

- 9 Cyanobacteria are aquatic blue-green bacteria which are highly similar to algae as they can obtain their energy through photosynthesis. They are typically found in tropical waters due to their ability to thrive in bright and warm areas. Generally found on the surface of lakes and oceans, they can reproduce exponentially and cause a rapid increase in their population known as blooms. Certain genera of blooming cyanobacteria such as *Microcystis* can produce cyanotoxins which, in high concentrations, can poison and even kill animals and humans.

A study was conducted to find out the effects of temperature on the maximum growth of *Microcystis aeruginosa* as well as three other harmless green algal species (P, Q and R) which are a main source of food for aquatic animals (Fig. 9.1). The global mean sea surface temperature anomalies, which indicate differences in temperature when compared to the baseline temperature in 1880 were also recorded (Fig. 9.2).

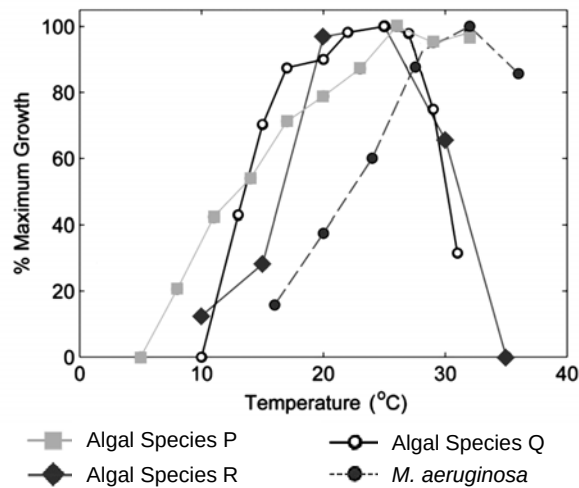


Fig. 9.1

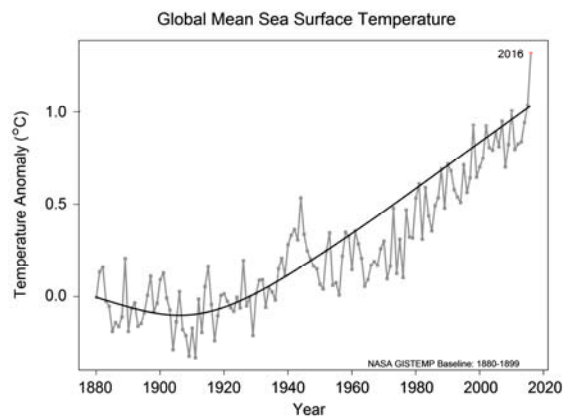


Fig. 9.2

- (a) Describe the effect of temperature on the maximum growth of *M. aeruginosa*.

[2]

- (b) Explain how human activities could have contributed to an increase in sea surface temperatures.

[3]

- (c) With reference to the information provided, discuss the impact of *M. aeruginosa* on the global food supply of humans in the future.

[5]

[Total: 10 m]

END OF PAPER

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Commented [LKM1]: Students must be able to identify different parts of the graph and explain them, while identifying trends (increasing trend)

Commented [LKM2]: Direct recall from LO – students have to phrase their answers in the context of SST

Commented [LKM3]: Similar to CA2 Q phrasing, where student can try to provide evidence where global food supply may be increased or decreased / assess validity of information in making such a claim